

United International University

Department of Computer Science and Engineering

Assignment: Software Engineering Laboratories (CSE 3422)

Section: A

Faculty: Shoib Ahmed Shourav

Assignment 1: Selenium Black Box Testing

Total Marks: 20

Submission Deadline: 23rd June, 2025

Objective

This assignment aims to introduce students to real-world black box testing using Selenium. Students will perform automated testing on a website (excluding `uiu.ac.bd`, `ucam.uiu.ac.bd`, and `elms.uiu.ac.bd`) using Selenium WebDriver in Python.

Tasks

1. Select a public website other than `uiu.ac.bd`, `ucam.uiu.ac.bd`, or `elms.uiu.ac.bd`.
2. Use Selenium WebDriver to:
 - Interact with at least two **text or search boxes**.
 - Test at least two **buttons**.
 - Test at least two **hyperlinks**.
3. Apply black box testing principles to verify proper functionality.
4. Record a **video** demonstrating your testing process and test results.
5. Prepare a **PDF report** including:
 - Website name and URL.
 - Test objectives and scenarios.
 - Your Python Selenium code with explanations.

- Screenshots of your test outputs.
- Summary of what you observed during testing.

6. Prepare another **separate PDF** answering the following question:

“What are the other available software testing tools? Which testing method (Black Box, Gray Box, or White Box) is most suitable for API testing and Database testing, and why?”

Submission Instructions

- Submit your assignment via the LMS.
- Submission files:
 - One video file demonstrating your Selenium test.
 - One PDF report of the testing details and code.
 - One PDF answering the theoretical question.
- Late submissions will be penalized.

Note: Academic integrity must be maintained. Any kind of plagiarism or duplication will result in mark deduction or disqualification.

— **End of Assignment** —